PROJECT REPORT

On

EXPLAINABLE DEEP LEARNING FRAMEWORK FOR EARLY RISK PREDICTION OF RETINOPATHY OF PREMATURITY USING EFFICIENTNET-B3 AND GRAD-CAM

A Project Report submitted to

Alliance University, Bengaluru

For the award of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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ALLIANCE UNIVERSITY - BENGALURU

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COMPUTER SCIENCE AND ENGINEERING

ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the project work entitled "Explainable Deep Learning Framework for Early Risk Prediction of Retinopathy of Prematurity Using EfficientNet-B3 and Grad-CAM" submitted by U. C Pravallika 2022BCSE07AED228, L B Vignesh 2022BCSE07AED283, Sinchana Suresh G 2022BCSE07AED129 and N Nanda Vardhan Reddy 2022BCSE07AED347 in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering(DA) of Alliance University, is a bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the project entitled **“Explainable Deep Learning Framework for Early Risk Prediction of Retinopathy of Prematurity Using EfficientNet-B3 and Grad-CAM”** submitted by us in the partialfulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE-DA) of ASAC, is a record of our work carried under the supervision and guidance of Dr. Ganga Holi, Professor, Computer Science and Engineering.

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty

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PREFACE

Retinopathy of Prematurity (ROP) continues to be a major cause of preventable blindness among premature infants, mostly because their retinal blood vessels don't fully develop. Catching ROP early is crucial—doctors can prevent permanent vision loss if they diagnose and treat it in time. But the standard diagnosis relies on 3D imaging and manual interpretation by ophthalmologists, a process that's often subjective, slow, and not always available in under-resourced clinics. This project introduces a machine learning–powered system to make ROP detection faster, more reliable, and easier to access, using only 2D retinal fundus images. We used deep learning models such as CNNs and segmentation methods like U-Net to automatically detect disease stages and highlight key clinical features like the demarcation line and ridge formations. To make the model robust, preprocessing steps—normalization, image enhancement, and data augmentation—were built in. We also developed a web-based tool where ophthalmologists can upload retinal images and get instant diagnostic predictions. The platform reports not only the likely diagnosis and a confidence score but also visual heatmaps that show exactly which image regions the model found important. This explainable AI (XAI) element is key—it builds trust and bridges the gap between automated tools and real clinical practice. Bringing automation together with transparency, our goal is to help standardize ROP diagnosis, reduce human error, and make early screening more widely available in neonatal intensive care units. In the end, this work supports faster intervention, better diagnostic accuracy, and improved visual outcomes for infants at risk, wherever they are in the world.

ABSTRACT

Retinopathy of Prematurity (ROP) is a retinal vascular disorder that threatens vision in premature infants due to incomplete blood vessel development. Precise staging is needed for timely treatment, but today’s standard methods rely on expensive 3D imaging and time-consuming manual review by ophthalmologists—a real problem in clinics with limited resources. On top of that, 2D neonatal eye images often suffer from poor quality, low resolution, and motion blur, complicating diagnosis even further. This project tackles these problems by proposing a machine learning model that approaches the diagnostic quality of 3D imaging, while operating directly on 2D retinal fundus photographs. Our model not only predicts ROP risk and staging but also marks out critical clinical features like the demarcation line and ridge formations. We leverage deep learning architectures—CNNs and segmentation models like U-Net-for both feature extraction and region localization. Robust preprocessing steps (image enhancement, normalization, data augmentation) help overcome issues with image quality and generalization. An interactive web interface lets ophthalmologists upload images and patient details, then receive immediate predictions—including the ROP stage, confidence scores, and visual explanations of which areas drove the model’s decision. By making use of explainable AI, we help build clinical trust in automation. This project closes the gap between affordable 2D imaging and labour-intensive 3D evaluations, making ROP diagnosis faster and more standardized, especially in neonatal intensive care units where staff and advanced equipment are scarce. Our system supports early detection, higher accuracy, and clear, stage-wise interpretation to better manage ROP in vulnerable infants.

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LIST OF ABBREVATIONS & SYMBOLS

Abbreviation / Symbol	Meaning
CNN	Convolution Neural Network
ROP	Retinopathy of Prematurity
Grad-CAM	Gradient-weighted Class Activation Mapping
DL	Deep Learning
ML	Machine Learning
XAI	Explainable Artificial Intelligence
NICU	Neonatal Intensive Care Unit
RGB	Red Green Blue
GPU	Graphics Processing Unit

CHAPTER 1
INTRODUCTION

Retinopathy of Prematurity (ROP) is an extreme and preventable eye disease that commonly influences premature and low birth-weight babies. It arises because of strange vascular improvement withinside the retina, which, if no longer diagnosed and dealt with early, can progress to retinal detachment and irreversible blindness. With the development in neonatal care, the survival rate of premature babies has expanded significantly, main to a developing range of ROP instances worldwide.

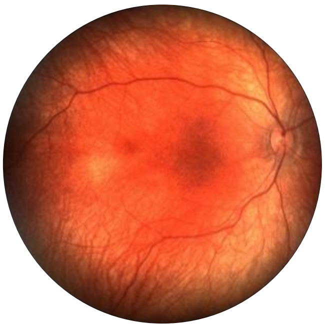

Figure 1.1: ROP

Traditionally, ophthalmologists diagnose and stage ROP using 3D retinal imaging systems, such as RetCam, which provide detailed depth information about the retina. However, these systems are expensive, not widely available in all hospitals, and require specialized expertise. In contrast, 2D retinal fundus images are easier to capture but lack the depth information necessary for precise staging. As a result, accurate diagnosis using 2D images remains a significant challenge, particularly in resource- limited settings.

Recent improvements in machine learning (ML) and deep learning (DL) have spread out new possibilities for automating clinical photo interpretation. In ROP detection, ML-based systems can learn to identify disease patterns and estimate severity levels directly from 2D images, potentially replicating or even enhancing the accuracy of 3D imaging systems. Deep learning architectures such as Convolutional Neural Networks (CNNs) and EfficientNet models are especially effective at extracting spatial and textural features that correspond to key diagnostic markers like demarcation lines, ridges, and vessel dilation.

1.1 Background

ROP is still a prime motive of preventable early life blindness, especially in premature infants. It results from incomplete vascularization of the retina, leading to abnormal vessel growth that can cause retinal detachment. The World Health Organization (WHO) highlights ROP as a critical neonatal condition requiring early detection and precise staging. However, in many developing countries, access to high-end 3D imaging systems and expert ophthalmologists is limited, creating disparities in screening quality and diagnosis accuracy. While 3D retinal imaging provides detailed depth information crucial for accurate ROP staging (refer figure 1.2), it is costly, time-intensive, and not feasible for large-scale or resource-limited screenings.

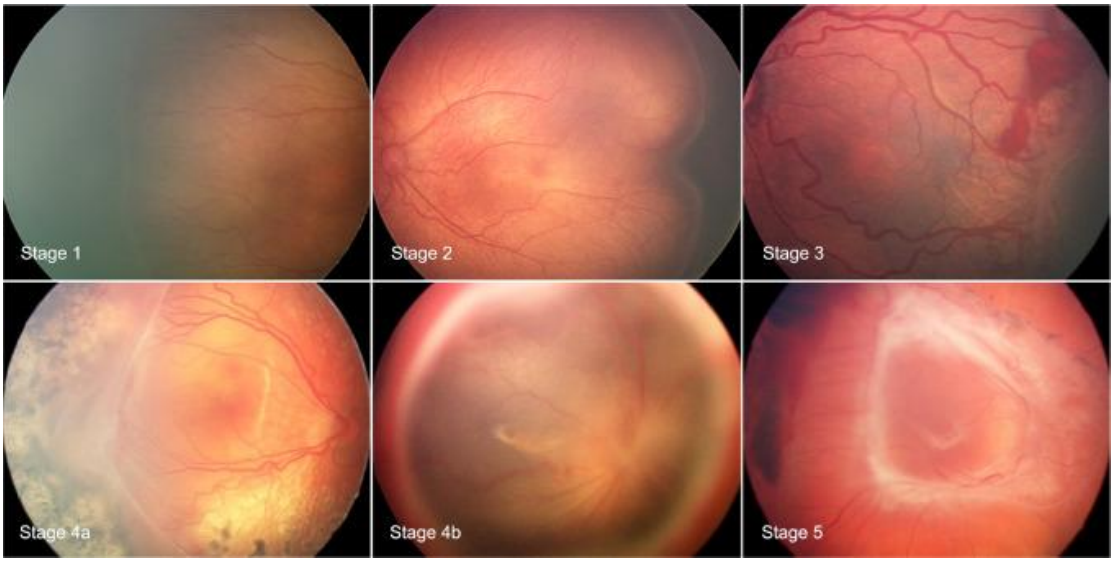

Figure 1.2: Stages of ROP

Conversely, 2D fundus images, though easier to acquire, lack depth perception, making it difficult to interpret retinal features like demarcation lines and ridges with precision. This gap motivates the need for a computational model that can replicate 3D-level diagnostic performance using 2D images. With rapid advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), there is now strong potential to overcome these limitations. CNN-based models have already demonstrated success in diagnosing diseases such as diabetic retinopathy and glaucoma. By applying similar methodologies to ROP, it is possible to extract subtle retinal patterns from 2D images and estimate disease stage accurately. The motivation behind this project is to create an automated, explainable, and clinically reliable system that can emulate ophthalmologists' diagnostic outcomes achieved with 3D imaging, using only 2D inputs.

1.1.1 Domain Overview

This project sits at the intersection of AI, ML, DL and Medical Image Processing. We specifically target healthcare—where machines assist human doctors in making diagnoses. Deep learning shines in medical imaging for its ability to sift through huge datasets and learn meaningful features, something traditional software struggles with. For retinal images, deep learning has already proven its value in detecting diabetic retinopathy, glaucoma, and more. Here, we adapt those techniques to the early detection of ROP. EfficientNet, particularly the B3 variant, stands out for balancing accuracy with efficiency—an important consideration in busy medical settings. Another core concern is explainability. Doctors need more than just a yes/no answer—they need insight into how an AI arrives at its conclusion. This is where XAI and techniques like Grad-CAM come in. Grad-CAM highlights the parts of an image most relevant to the prediction, supporting clinical interpretation and building trust. By combining deep learning with explainability, we build not just an accurate system, but a usable one, tailored to real-world demands of ROP screening.

1.2 Problem Statement

ROP is a severe retinal disorder that affects premature infants due to incomplete development of retinal blood vessels. If not diagnosed and treated at an early stage, ROP can rapidly progress to retinal detachment and permanent blindness. Early and accurate stage-wise diagnosis is therefore critical for timely intervention and effective treatment planning.

However, existing diagnostic approaches face several challenges. Traditional screening relies heavily on experienced ophthalmologists and advanced imaging systems such as RetCam, which are expensive and not readily available in low-resource healthcare settings. Additionally, manual diagnosis is time-consuming (refer figure 1.3), subjective, and prone to variability among experts. Many existing AI-based systems focus only on binary classification (ROP vs non-ROP) and lack precise stage-wise detection, interpretability, and real-time usability.

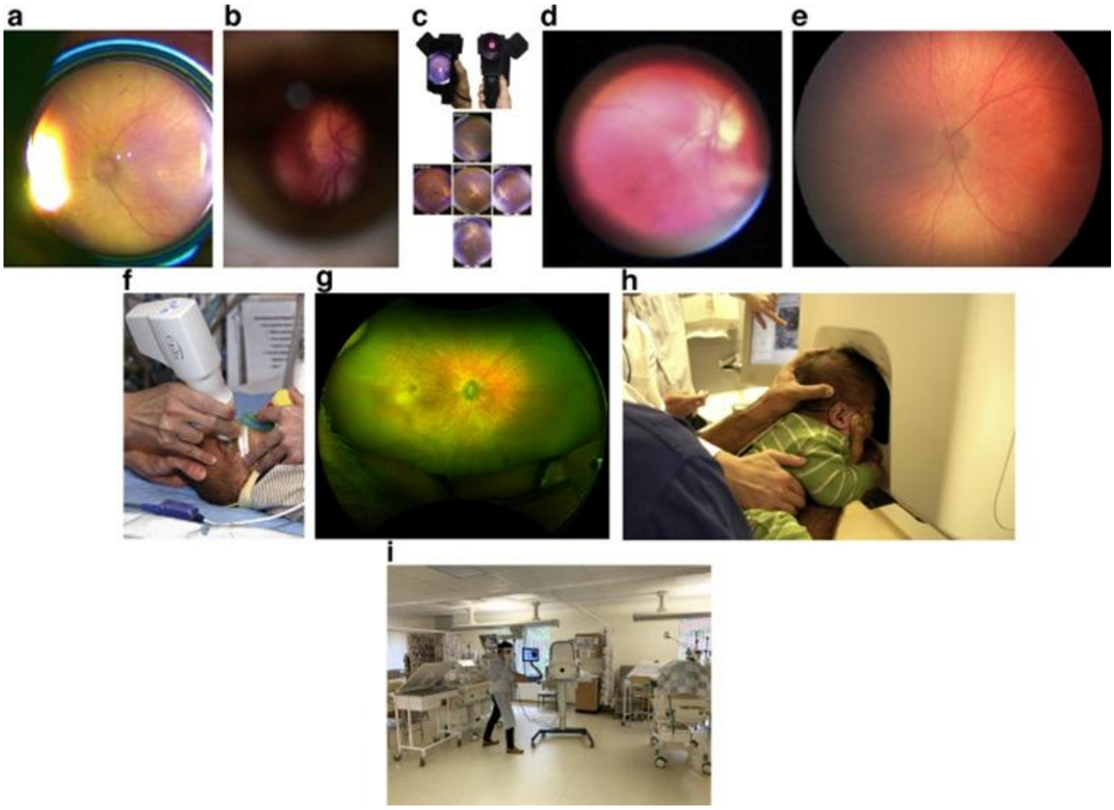

Figure 1.3: ROP screening

Furthermore, neonatal retinal images often suffer from poor quality due to low illumination, motion blur, and noise, making accurate analysis more difficult. Existing models also struggle with limited datasets, lack of generalization, and insufficient integration of clinical insights, reducing their reliability in real-world applications. Therefore, there is a critical need for a cost-effective, accurate, and explainable automated system that can detect and classify ROP using easily available 2D retinal fundus images, provide stage-wise predictions, and assist clinicians with interpretable visual insights for better decision-making.

1.3 Objectives

The primary objective of this project is to develop an intelligent and automated system for the detection of ROP using deep learning techniques. The specific objectives are as follows:

- To develop an automated ROP detection system: Design a machine learning model capable of identifying whether a retinal fundus image indicates the presence of ROP or not.
- To implement an effective image preprocessing pipeline: Enhance low-quality neonatal retinal images using techniques such as CLAHE, Gaussian blur, and normalization to improve feature visibility and model performance.
- To build a deep learning-based classification model: Utilize an EfficientNet-based Convolutional Neural Network (CNN) to extract features and perform accurate classification of retinal images.
- To provide explainable predictions using Grad-CAM: Generate visual heatmaps highlighting important regions in the retinal image to improve interpretability and clinical trust.
- To develop a user-friendly web interface: Create a Flask-based web application that allows users to upload retinal images and receive real-time predictions along with visual explanations.
- To generate automated diagnostic reports: Produce downloadable PDF reports containing prediction results, confidence scores, and Grad-CAM visualizations for clinical use.
- To maintain prediction history for analysis: Store previous predictions and results in a structured format to enable tracking and future reference.
- To ensure cost-effective and accessible screening: Provide a solution that works with 2D retinal images, reducing dependency on expensive 3D imaging systems and enabling deployment in low-resource settings.

1.4 Scope

The scope of this project is focused on the development of an intelligent, automated, and cost-effective system for the detection of Retinopathy of Prematurity (ROP) using deep learning techniques applied to 2D retinal fundus images. The system is designed to assist ophthalmologists and healthcare professionals in early screening and diagnosis, particularly in resource-constrained environments.

The proposed system covers the following aspects:

1. Automated ROP Detection: The system is capable of analyzing retinal fundus images and determining whether the infant is affected by ROP or not, reducing dependency on manual diagnosis.
2. Image Enhancement and Preprocessing: The project includes preprocessing techniques such as Gaussian blurring, CLAHE (Contrast Limited Adaptive Histogram Equalization), and normalization to improve the quality of retinal images and enhance important features.
3. Deep Learning-Based Classification: A CNN based on EfficientNet architecture is implemented to perform accurate classification of retinal images into ROP and non-ROP categories.
4. Explainable AI Integration: The system incorporates Grad-CAM visualization to highlight the regions of the retina that influence the model's prediction, improving interpretability and clinical trust.
5. Web-Based User Interface: A Flask-based web application is developed to provide a user-friendly platform where users can upload images, view predictions, and access results in real time.
6. Automated Report Generation: The system generates downloadable PDF reports containing prediction results, confidence scores, and visual explanations, making it suitable for clinical documentation.
7. Prediction History Management: The application maintains a record of previous predictions using a CSV-based storage system, enabling tracking and analysis of results.
8. Applicability in Low-Resource Settings: By relying on 2D retinal images instead of expensive 3D imaging systems, the solution is designed to be accessible and deployable in hospitals and clinics with limited resources.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing System

Existing systems for the detection and diagnosis of ROP primarily rely on traditional clinical examination methods and recently developed AI-based approaches. In conventional practice, ophthalmologists use specialized imaging devices such as RetCam to capture high-resolution retinal images for diagnosis. These systems provide detailed visualization of retinal structures; however, they are expensive, require expert handling, and are not widely available in low-resource healthcare settings. Additionally, manual diagnosis is time-consuming and subject to inter-observer variability, which may lead to inconsistencies in disease identification and staging.

With advancements in machine learning and deep learning, several automated systems have been proposed for ROP detection. Most of these systems utilize CNNs and other deep learning architectures such as ResNet, EfficientNet, U-Net, and Vision Transformers to analyze retinal images. These models have demonstrated high accuracy in detecting ROP and related features such as blood vessel abnormalities, demarcation lines, and ridge formations.

Some existing systems focus on binary classification, distinguishing between ROP and non-ROP cases, while others attempt segmentation and feature extraction to identify important retinal structures. Advanced approaches integrate techniques like Mask R-CNN for object detection and U-Net for vessel segmentation. Additionally, certain studies incorporate clinical data such as gestational age and birth weight to improve prediction accuracy.

Despite these advancements, many systems remain limited in practical deployment. Several models depend on high-quality retinal images and controlled datasets, which restrict their performance in real-world scenarios. Moreover, many systems lack explainability, making it difficult for clinicians to trust automated predictions. Only a few studies incorporate visualization techniques such as Grad-CAM to provide insights into model decisions.

Furthermore, most existing systems are not integrated into user-friendly applications and lack real-time processing capabilities. They often remain at the experimental or research level, without practical deployment in clinical environments.

In summary, while existing systems have shown promising results in automating ROP detection, they face challenges related to cost, accessibility, interpretability, and real-world applicability, highlighting the need for more robust and deployable solutions.

1.2 Research Review

The paper titled "Digital Enhancement of Neonatal Fundus Color Photography: Improving RoP Diagnosis through Image Processing" by S. Salahi-Mogaddam et al. [1] (2024 IEEE ICBME Conference) focuses on improving the quality of neonatal retinal images for more accurate diagnosis of ROP. The authors emphasize that poor illumination, motion blur, and low contrast in neonatal fundus photographs often lead to diagnostic errors and delay in treatment. To address these challenges, the study evaluates two digital enhancement techniques — Contrast Limited Adaptive Histogram Equalization (CLAHE) and Automated Multi-Scale Retinex with Color Restoration (Automated MSRCR) — applied to fundus images captured using RetCam3 at Farabi Eye Hospital, Tehran. These methods were assessed qualitatively by ophthalmologists based on image clarity, visibility of retinal structures, and diagnostic usefulness. Experimental results showed that both techniques significantly enhanced image quality, especially for low-quality images, with Automated MSRCR providing better illumination balance and color fidelity. The researchers also developed a Python-based graphical user interface (GUI) using PySimpleGUI to simplify clinical implementation. However, the study's limitations include a small dataset (24 images), reliance on subjective expert grading instead of quantitative metrics, and the absence of automated ROP stage classification. The authors suggest that future work could integrate these enhanced images into AI-driven diagnostic models for real-time screening and improved clinical decision-making.

The paper titled "A Study on Retinopathy of Prematurity (ROP) Screening Using Deep Learning Approaches and a Blood Vessel Segmentation Framework for ROP Affected Eyes" by Madhu K., et al. [2] (2024, ICETITE Conference) presents a comprehensive review of deep learning methods for ROP detection, classification, and segmentation.

The authors analyzed 22 studies (2016–2023) employing CNN-based models such as ResNet, VGG, YOLO, and U-Net, applied to private and public retinal image datasets. They identified major challenges like low-quality neonatal images, lack of annotated public datasets, and multi-stage ROP presence in single images, which complicate diagnosis. After the survey, the authors proposed a segmentation framework (ROP-Seg) combining U-Net and EfficientNet architectures for blood vessel segmentation using a publicly available dataset of 101 fundus images. The model achieved 84.5% validation accuracy, demonstrating potential for improved vascular analysis in premature infants. However, the paper notes limitations such as low intersection-over-union (IoU) scores, small dataset size, and absence of lesion localization or zone detection. The authors suggest that future research should integrate multi-task learning models capable of stage detection, zone classification, and explainable AI for better clinical reliability.

The paper titled "Retinopathy of Prematurity Stage Diagnosis Using Object Segmentation and Convolutional Neural Networks" by Alexander Ding et al. [3] (IEEE, 2020) proposes a hybrid deep learning architecture that integrates object segmentation with Convolutional Neural Networks (CNNs) to classify ROP stages 1–3 from neonatal retinal images. The authors emphasize that accurate staging depends on detecting the demarcation line, a key retinal feature distinguishing early ROP stages. Their method first employs a Mask R-CNN model to segment demarcation lines and then feeds the resulting binary mask along with the original preprocessed image into an InceptionV3 CNN classifier for improved stage prediction. Using transfer learning and data augmentation, the model achieved 67% classification accuracy, outperforming CNN-only and segmentation-only approaches by 13% and 20%, respectively. However, the study faced limitations such as imbalanced datasets, single-expert annotations, and modest F1-scores, especially between Stages 2 and 3. The authors recommend expanding datasets, including ROP-free images, and combining their segmentation approach with multi-instance learning for enhanced clinical applicability.

The paper titled "Development of an Automated Screening System for Retinopathy of Prematurity Using Deep Convolutional Neural Network for Wide-Angle Retinal Images" by Fahimeh Rabiei et al. [4] (2023, Elsevier Biomedical Signal Processing and Control) presents an automated deep learning–based screening framework for detecting ROP from wide-field retinal images. The authors employ a Deep Convolutional Neural Network (DCNN) trained on RetCam images to classify cases into normal and ROP-affected categories. The model architecture incorporates multiple convolutional and pooling layers for feature extraction and a softmax classifier for binary decision-making. The proposed system achieved 95.8% accuracy, 96% sensitivity, and 95% specificity, demonstrating strong performance compared to conventional image-based diagnostic methods. The study also includes preprocessing steps such as noise reduction, resizing, and contrast normalization to improve image quality. Despite high accuracy, the system has certain limitations: it performs only binary classification (normal vs. ROP) without stage-wise diagnosis, lacks explainability mechanisms, and depends heavily on a limited dataset with no cross-hospital validation. The authors recommend integrating transfer learning and expanding datasets for robust and generalizable ROP screening in clinical applications.

The paper titled "Prediction Models for Retinopathy of Prematurity Occurrence Based on Artificial Neural Network" by Rong Wu et al. [5] (BMC Ophthalmology, 2024) presents a noninvasive predictive model for identifying ROP risk using Back Propagation (BP) Neural Network and Genetic Algorithm–Back Propagation (GA-BP) Neural Network. The study analyzed data from 591 preterm infants, evaluating maternal, neonatal, and intervention-related factors to determine key predictors such as gestational age, birth weight, mechanical ventilation, and blood transfusions. The GA-BP model outperformed both logistic regression and ROP score models, achieving an AUC of 0.908, sensitivity of 92.1%, and accuracy of 90.1%. Results demonstrated the superiority of GA-BP networks in handling non-linear clinical data for early ROP prediction. However, limitations include a small dataset, lack of stage-wise classification, and the "black-box" nature of neural networks, which reduces interpretability. The authors suggest further validation with larger datasets and incorporation of explainable AI for improved clinical trust and deployment.

The paper titled "Deep Learning–Based Automated Screening of Retinopathy of Prematurity Using Fundus Images" by Jing Wang et al. [6] (Nature Scientific Reports, 2023) proposes an end-to-end deep learning framework for automated ROP detection and classification using neonatal wide-field fundus images. The authors employ a ResNet-50 backbone CNN integrated with a feature attention mechanism to classify disease stages and identify treatment-requiring cases. The system was trained on 5,250 labelled retinal images collected from multiple neonatal units and achieved an AUC of 0.953, outperforming traditional CNN models. Visualization tools like Grad-CAM were used to highlight disease-relevant retinal regions, enhancing interpretability. The paper also emphasizes the importance of cross-hospital generalization by validating the model on an external dataset. However, the study is limited by imbalanced class distribution, lack of segmentation for the demarcation line or ridge, and dependence on high-quality imaging devices. The authors recommend future work on multi- task models that integrate segmentation and clinical metadata for comprehensive ROP diagnosis.

The paper titled "Early Detection of Retinopathy of Prematurity Stage Using Deep Learning Approach" by Supriti Mulay et al. [7] (arXiv, 2021) presents a deep learning–based framework for ridge and demarcation line detection in early stages of Retinopathy of Prematurity (ROP). The authors emphasize that identifying the ridge — the key clinical landmark distinguishing Stages 1 and 2 — is vital for early intervention. The proposed model utilizes a Mask R-CNN architecture with a ResNet-101 backbone to detect and segment ridge regions in neonatal retinal images from the KIDROP dataset (220 images from 45 infants). Image enhancement through Modified Histogram Equalization (MHE) and CLAHE in YIQ color space was performed to address poor illumination and low contrast. The model achieved a ridge detection accuracy of 0.88, with a precision of 0.97, recall of 0.90, and F1-score of 0.93, outperforming traditional image processing methods. The study demonstrated that preprocessing significantly improved detection performance, as raw images often failed to localize ridges accurately. However, limitations include a small dataset, difficulty distinguishing between Stage 1 and Stage 2, and reliance on manual annotations. The authors recommend expanding the dataset and extending the approach for full stage - wise ROP classification and segmentation-based clinical applications.

The paper titled "Automated Detection and Stage Classification of Retinopathy of Prematurity Using Deep Learning Techniques" by Lopa Mudra and Nirmalya Kar [8] (International Journal of Computer Science and Information Security, 2023) proposes a deep learning–based automated diagnosis system for ROP detection and stage identification from neonatal fundus images. The study uses a two- stage pipeline: a CNN-based classifier for determining ROP severity and a segmentation network (U-Net) to highlight abnormal vascular structures to assist clinical interpretation. The dataset comprises RetCam-wide field images, which undergo preprocessing including noise removal, resizing, and contrast enhancement. The model achieved an overall accuracy of 92.5% in differentiating normal, mild, and severe ROP cases. The authors highlight that segmentation improves explainability by visualizing pathological features such as vessel tortuosity. However, limitations include small dataset availability, difficulty discriminating between adjacent stages (Stage 2 vs Stage 3), and reliance on manual expert annotations for training. Further improvements such as transfer learning, data augmentation, and multi-center validation are suggested to enhance the model's clinical deployment potential.

The paper titled "Development and Validation of Machine Learning Classifiers for Predicting Treatment-Needed Retinopathy of Prematurity" by Nasser Shoeibi, Majid Abrishami et al [9]. (2025, BMC Medical Informatics and Decision Making) presents a comparative evaluation of multiple supervised machine-learning algorithms for identifying premature infants who require treatment for ROP using demographic and clinical data rather than retinal images. The study analyzed medical records of 9,692 infants and considered 12 key features, including gestational age, birth weight, NICU duration, ROP stage, and zone. Eight classifiers—Logistic Regression, Decision Tree, SVM, Naïve Bayes, KNN, XGBoost, ANN, and Random Forest—were trained and validated using 5-fold cross-validation and oversampling techniques (SMOTE and ADASYN) to address data imbalance. Results showed that XGBoost and ANN achieved the highest accuracy of 96%, while Naïve Bayes provided the best sensitivity (0.99), reducing false negatives—crucial for clinical applications. The study also applied SHAP analysis for explainability, identifying NICU stay, birth weight, and ROP stage as key predictive factors. However, the authors note limitations such as reliance on single-center data, absence of imaging or oxygen-therapy features, and lack of external validation. They emphasize that future work should integrate multicenter datasets and deep learning models for broader generalizability. The research demonstrates that ML models can effectively support ophthalmologists in early ROP risk assessment but should complement, not replace, clinical judgment.

The paper titled "Detection of Plus Disease in Retinopathy of Prematurity Using Deep Learning Models: Evaluating Vision Transformers and ResNet Architectures" by Ibrahim Kocak, Sadık Etka Bayramoglu et al. [10] (2025, International Journal of Imaging Systems and Technology) investigates automated detection of Plus Disease (PD) — a critical indicator of severe Retinopathy of Prematurity (ROP) — using Vision Transformer (ViT) and ResNet 50 architectures. The study utilized 1,205 fundus color images from the Kanuni Sultan Suleyman Research Hospital, Turkiye, and generated corresponding vascular segmented mask images using a U Net model. Both ViT and ResNet-50 were trained and tested on color and segmented datasets, followed by external validation with a public dataset. Results showed that ViT achieved higher accuracy (96.9%) and F1-score (96.9%) on color images, while ResNet-50 demonstrated superior recall (100%) and efficiency, making it more suitable for clinical screening where missing cases is critical. Although segmentation was expected to improve performance, both models performed better on raw color images, indicating limited benefit from vascular masking. The study concludes that ResNet-50 is an optimal choice for PD detection due to its strong generalization, shorter training time, and perfect recall rate, while ViT offers potential advantages for larger datasets. Limitations include single-center data, exclusion of pre-plus stage, and small dataset size, which may affect generalizability. Future work should explore hybrid CNN–Transformer architectures and multi-stage classification to enhance the accuracy and clinical applicability of AI- based ROP screening systems.

The paper titled "Retinal Image Dataset of Infants and Retinopathy of Prematurity" by Juraj Timkovič, Jana Nowaková et al. [11] (2024, Scientific Data, Nature Publishing Group) introduces a publicly available dataset of 6,004 retinal fundus images from 188 infants, most of whom are premature, collected at the University Hospital Ostrava, Czech Republic. The dataset includes labeled clinical data such as gestational age, birth weight, postconceptual age, diagnosis, and plus disease status, supporting research in automated ROP detection and computer-aided diagnosis. The authors also developed a software tool named ReLeSeT, designed for automatic retinal lesion segmentation using an active contour model without edges, combined with bilateral filtering and adaptive histogram equalization for noise reduction and feature enhancement. The tool quantifies lesion geometry and intensity, enabling feature extraction for subsequent ML analysis. Evaluation using ground-truth expert annotations achieved ~99% accuracy and specificity, confirming strong segmentation reliability across three imaging systems (Clarity RetCam 3, Natus RetCam Envision, and Phoenix ICON). Despite its strengths, the paper notes some limitations—such as variable image quality due to infant non-cooperation, limited representation of advanced ROP stages (4A–5), and device-dependent intensity variations. The dataset is considered a valuable resource for developing deep learning models for ROP stage classification and vascular lesion quantification, paving the way for improved clinical screening and automated ROP assessment.

The paper titled "Synthetic Medical Images for Robust, Privacy-Preserving Training of Artificial Intelligence: Application to Retinopathy of Prematurity Diagnosis" by Aaron S. Coyner, Jimmy S. Chen, et al [12] (2022, Ophthalmology Science) explores the use of progressively growing generative adversarial networks (PGANs) to generate synthetic retinal vessel maps for detecting plus disease in ROP while maintaining patient privacy. Using 5,842 retinal fundus images from 963 preterm infants, vessel maps were segmented via U-Net, and PGANs were trained to synthesize normal, pre-plus, and plus-disease vasculatures. Two ResNet-18 CNNs were trained— one on real data and one on synthetic data—and evaluated using AUC, UMAP visualization, Fréchet Inception Distance (FID), and Euclidean similarity measures. The CNN trained on synthetic data achieved AUC = 0.971, outperforming the real-data model (AUC = 0.934), and demonstrated near-perfect agreement (κ = 0.922) with eight international ROP experts. Analyses confirmed that synthetic images were not replicas of real patient data but captured realistic disease patterns, enabling privacy-preserving dataset sharing. Despite these advantages, the study acknowledges limitations, including use of a single consortium dataset, absence of formal privacy-preserving constraints during GAN training, and restriction to grayscale vessel maps. The authors conclude that synthetic medical images can enhance model robustness, reduce data-sharing barriers, and expand ROP AI research without compromising patient confidentiality.

The paper titled "Artificial Intelligence for the Diagnosis of Retinopathy of Prematurity: A Systematic Review of Current Algorithms" by Ashwin Ramanathan, et al [13] (2023, Eye, Nature Publishing Group) provides a systematic review of 27 studies utilizing deep learning (DL) algorithms for the detection and classification of ROP. The review summarizes models trained on wide-field digital retinal images (WFDRI) for tasks such as detecting ROP versus no ROP, identifying plus/pre-plus disease, assessing image quality, and classifying disease stage or severity. Reported performances across studies showed sensitivities of 71–100%, specificities of 74–99%, and AUROC values of 0.91–0.99, indicating diagnostic accuracy comparable to ophthalmologists. The review highlights that many algorithms employed CNN architectures (e.g., ResNet, VGG, Inception, U-Net) and some used vascular severity scores (VSS) to quantify disease progression. However, most studies were retrospective, used high-quality archived images, and lacked real-world clinical validation. Only two evaluated AI tools prospectively in hospital settings. Key limitations identified include dataset heterogeneity, lack of standardized imaging protocols, and limited generalizability to low-resource regions. The authors emphasize that while AI can serve as an adjunct diagnostic aid to reduce ophthalmologists' workload, it is not yet ready to replace human evaluation. They recommend large-scale, multicenter clinical validation and stronger ethical and governance frameworks before full integration into neonatal screening programs.

The paper titled "Automated Detection of Stages, Zones, and Plus Diseases of Retinopathy of Prematurity Using Quantum Convolutional Networks in Neonatal Fundus Images" by Raja Sankari V.M., Snekhalatha Umapathy et al. [14] (2025, Engineering Applications of Artificial Intelligence) introduces a Quantum Mobile Vision Transformer (QMViT) that integrates quantum computing, vision transformers, and pre-trained CNNs for multi- class classification of ROP stages, zones, and plus disease. The model leverages a hybrid architecture combining Swin UNet for blood vessel segmentation and quantum-enhanced transformer layers for classification. The study trained on 2,400 retinal images (2,000 ROP and 400 normal), using Swin UNet and MultiResUNet for segmentation, followed by feature extraction (SIFT, SURF, ORB), PCA- based dimensionality reduction, and classification with SVM, Random Forest, and k-NN. For deep learning, QMViT was compared against ViT, Swin Transformer, ResViT, and MobileViT. Results showed QMViT achieved accuracies of 95.5%, 96.88%, and 96.67% in classifying ROP stages, zones, and plus disease, respectively, outperforming all baseline models. The quantum-enhanced model demonstrated computational efficiency through a 4-qubit quanvolutional layer, allowing localized nonlinear transformations and improved feature representation. Despite high performance, the authors note limitations such as a small dataset, moderate overfitting in some tasks, and lack of external validation. The study suggests future improvements through larger multi-institutional datasets, real quantum hardware implementation, and integration of clinical metadata to enhance robustness and clinical readiness.

The paper titled "Deep Learning for the Diagnosis of Stage in Retinopathy of Prematurity: Accuracy and Generalizability across Populations and Cameras" by Jimmy S. Chen, Aaron S. Coyner, et al. [15] (2021, Ophthalmology Retina) presents a

CNN-based system for detecting stages 1–3 of ROP and evaluating its generalizability across populations and imaging devices. The study used 5,943 RetCam images from North America and 5,049 3nethra images from Nepal, each labeled by expert graders. A ResNet-152 model pretrained on ImageNet was fine-tuned with 5-fold cross-validation and evaluated separately on each population and a combined dataset. The North American-trained model achieved AUROC = 0.99, AUPRC = 0.98, and 94% sensitivity on internal data but dropped to AUROC = 0.96 and 52% sensitivity on Nepali data. Conversely, the Nepali-trained model scored AUROC = 0.97 internally but only 0.62 on the North American dataset, showing a significant domain shift between populations. The combined model improved generalization, achieving 98% sensitivity (North America) and 82% sensitivity (Nepal). The paper demonstrates that heterogeneous, multi-population training datasets enhance external performance and robustness of AI models. However, limitations include dataset imbalance, preprocessing differences, and the lack of multiclass explainability or integration of other ROP features (zone and plus disease). Future work should involve cross institutional validation and adaptive domain-transfer learning to achieve clinically reliable, camera independent ROP screening.

The paper titled "Advances in Early Detection and Monitoring of Retinopathy of Prematurity in Preterm Infants Using CNN and MLP Models" by G. Hubert, S. Silvia [16] Priscila presents a hybrid deep learning framework combining Convolutional Neural Networks (CNNs) and Multilayer Perceptron (MLP) models for early detection and continuous monitoring of Retinopathy of Prematurity (ROP) progression. The authors utilized a dataset of 3,200 preprocessed fundus images acquired from neonatal intensive care units (NICUs) and paired these with clinical variables such as gestational age, birth weight, and oxygen therapy duration. The CNN component extracted spatial and structural retinal features, while the MLP integrated these image-based features with clinical metadata for multimodal prediction. The hybrid CNN–MLP architecture achieved an overall accuracy of 96.2%, sensitivity of 95.6%, and specificity of 97.1%, outperforming single-modality models. The inclusion of clinical data improved prediction robustness and reduced misclassification between adjacent ROP stages. However, the study acknowledges limitations, including a modest dataset size, limited image diversity, and lack of external validation. The authors suggest expanding training data using multi center collaboration and incorporating temporal progression modeling for longitudinal ROP monitoring. The study demonstrates that integrating clinical parameters with imaging data through deep learning can significantly enhance ROP stage prediction and early intervention potential.

The paper titled "Global Prevalence and Severity of Retinopathy of Prematurity over the Last Four Decades (1985–2021): A Systematic Review and Meta-Analysis" by Heladia García, Miguel Angel Villasis-Keever, et al. [17] (Archives of Medical Research, 2024) performs a systematic review and meta- analysis of 139 studies (121,618 premature infants) to estimate worldwide ROP prevalence and severity trends from 1985–2021. The authors searched PubMed, Embase, and Google Scholar, applied standard inclusion criteria, extracted clinical/demographic data, and pooled prevalence using random-effects models in R; subgroup analyses were performed by gestational age, decade, world region, neonatal mortality rate, and country income. The pooled prevalence of any ROP was 31.9% (95% CI 29.0–34.8) and severe ROP 7.5% (95% CI 6.5–8.7); highest prevalence occurred in infants ≤28 weeks GA and in lower-middle-income countries, while severe ROP was comparatively elevated in high-income settings. The study emphasizes persistent global ROP burden and the need for targeted screening in high-risk populations. Key limitations reported are high heterogeneity across studies, variable image/diagnosis criteria, under-representation of advanced stages, reliance on published (often single-center) datasets, and limited reporting of plus disease or standardized grading, all of which may affect pooled estimates and generalizability.

The paper titled "A Fundus Image Dataset for Intelligent Retinopathy of Prematurity System" by Xinyu Zhao, Shaobin Chen, et al [18] (2024, Scientific Data, Nature Publishing Group) presents a large-scale ROP fundus image dataset designed to support the development and validation of AI-based diagnostic systems for neonatal eye disease. The dataset comprises 1,099 wide-field fundus images from 483 preterm infants collected at Shenzhen Eye Hospital, China, between 2004 and 2023, including images classified as Normal, Stage 1, Stage 2, Stage 3, and Laser scars. Images were captured using multiple devices (RetCam, SW-8000, Nautilus) and underwent uniform preprocessing (512×512 resolution cropping, quality control, and illumination normalization). Classification followed the ICROP-3 criteria and was performed independently by two junior annotators, with a senior ophthalmologist resolving discrepancies (Cohen's κ = 0.856). To validate dataset quality, several deep learning models — ResNet50, ResNet101, ConvNeXt-T, and ViT-B — were trained; ResNet50 achieved the best performance with accuracy = 87.6% and AUC = 96.7%. The dataset demonstrates strong annotation consistency and technical validity but remains limited to Asian infants and excludes advanced stages (4–5). The authors emphasize that this open-access dataset will facilitate multi-center AI research, segmentation-based lesion analysis, and stage-wise ROP classification in real-world clinical applications.

The paper titled "Automated Classification of Prematurity Retinopathy Using ViT, ResNet, and EfficientNet" by A. Pérez, D. Karpov et al. [19] (2025, IEEE Ural Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology – USBEREIT) presents a comparative study of Vision Transformer (ViT) and Convolutional Neural Network (CNN) architectures for automated classification of Retinopathy of Prematurity (ROP). The research focuses on identifying optimal model configurations for detecting and classifying ROP severity stages from neonatal retinal fundus images. The study utilized multiple pre-trained architectures — ViT-B/16, ResNet-50, and EfficientNet-B4 — fine-tuned on a labeled dataset of fundus images preprocessed with CLAHE, Gaussian blurring, and normalization to enhance vessel and demarcation line visibility. Among all tested models, EfficientNet-B4 achieved the best overall performance with an accuracy of 97.4% and F1-score of 96.8%, outperforming both ViT and ResNet 50 in balancing precision and recall. The study also highlights the efficiency of transformer-based models in capturing global contextual retinal features compared to CNNs, which focus on local patterns. However, the authors acknowledge dataset limitations, particularly small sample size and lack of cross-hospital validation, which may hinder generalization. They recommend combining CNN and Transformer features in hybrid frameworks and integrating explainability tools (e.g., Grad-CAM) for better clinical interpretability. The paper demonstrates that lightweight transformer models can achieve high diagnostic accuracy for ROP detection, offering potential for real-time neonatal screening systems.

The paper titled "Prediction of Retinopathy of Prematurity Development and Treatment Need with Machine Learning Models" by Ceren Durmaz Engin et al. [20] (2025, BMC Ophthalmology) investigates the potential of machine learning (ML) algorithms to predict the development of retinopathy of prematurity (ROP) and the need for treatment in preterm infants. The study used data from 355 preterm infants and developed four ML models based on 49 clinical, maternal, and neonatal parameters collected within 24 hours after birth and during the first ROP screening. The authors employed five classifiers — Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost) —and evaluated their performance using metrics like sensitivity, specificity, balanced accuracy, and F1 score. Among the models, XGBoost and LR achieved the best results, with balanced accuracies of around 80–81% for predicting ROP and treatment needs. The most influential features were gestational age, birth weight, mean corpuscular volume (MCV), APGAR scores, oxygen support duration, and daily weight gain. The study used SMOTE to handle data imbalance and 5-fold cross-validation for validation. However, the authors noted several limitations, including the small dataset, single-center data source, class imbalance (only 12.7% required treatment), and lack of external validation or fundus imaging data. Despite these drawbacks, the study demonstrates that integrating clinical and intensive care parameters through ML models can effectively assist in early ROP risk prediction and clinical decision-making, particularly in settings without imaging facilities.

The paper titled "Early Diagnosis and Quantitative Analysis of Stages in Retinopathy of Prematurity Based on Deep Convolutional Neural Networks" by Peng Li and Jia Liu (2022, Translational Vision Science & Technology, Vol. 11, No. 5) [21] proposes a fully automated DCNN-based system for early-stage ROP detection and quantitative feature analysis. The study utilizes 18,827 retinal images from preterm infants collected at Jiaxing Maternal and Child Health Hospital, China, focusing on diagnosing normal, Stage I, Stage II, and Stage III ROP. The system integrates two Retina U-Net models for segmenting blood vessels and demarcation lines/ridges, followed by DenseNet for classification. Quantitative parameters such as demarcation line width and vascular proliferation ratio within the region of interest (ROI) were computed to assist clinical decision-making. The model achieved 95.33% overall accuracy, with sensitivities and specificities exceeding 90% for each stage (e.g., Stage I: 90.21%/97.67%, Stage II: 92.75%/98.74%, Stage III: 91.84%/99.29%). The study also introduced a quantitative interpretability layer, demonstrating that wider demarcation lines and higher vascular proliferation ratios correlate with higher ROP stages, improving physician diagnostic consistency ($\kappa = 0.9425$). However, limitations include the use of only high-quality RetCam3 images, single-center data, and exclusion of Stages IV–V ROP. The authors suggest expanding data diversity, incorporating multi-device imaging, and improving explainability visualization in future work.

The paper titled “Classification of Stages 1, 2, 3 and Pre-plus, Plus Disease of ROP Using MultiCNN_LSTM Classifier” by Ranjana Agrawal, Sucheta Kulkarni et al. [22] (2025, MethodsX, Elsevier) presents a hybrid MultiCNN_LSTM deep learning model for automated classification of Retinopathy of Prematurity (ROP) stages (1–3) and identification of Pre-plus/Plus disease from fundus images. The study utilizes two datasets—HVDROPDB-STAGE and HVDROPDB-PLUS—comprising RetCam and Neo fundus images. Ridge segmentation was performed using Attention-Gate U-Net, and cropped ridge patches were used for detailed stage-wise classification. Features were extracted using three pretrained CNN architectures—InceptionV3, Xception, and ResNet152—and fused before being passed into a Long Short-Term Memory (LSTM) classifier. The proposed MultiCNN_LSTM model achieved 93% accuracy for stage classification and 87% accuracy for Pre-plus/Plus classification, outperforming individual CNNs and CNN-LSTM models in both accuracy and F1-scores. The study demonstrates that combining multiple CNN feature extractors with an LSTM classifier significantly improves the detection of subtle stage transitions and vascular abnormalities. However, the authors note limitations including single-region dataset collection (Indian population), small dataset size, and absence of external validation. Future work aims to expand the dataset, integrate zone detection, and develop explainable AI visualizations to enhance clinical interpretability and scalability of ROP diagnosis systems.

The paper titled “Automated Detection of Type 1 ROP, Type 2 ROP, and A-ROP Based on Deep Learning” by Eşay Kıran Yenice, Caner Kara et al. [23] (2024, Eye, Nature Publishing Group) introduces a CNN-based deep learning model for automatically classifying Retinopathy of Prematurity (ROP) into Type 1, Type 2, and Aggressive ROP (A-ROP) using clinical fundus images. A dataset of 634 fundus images from 317 premature infants (23–34 weeks GA) was preprocessed using Adaptive Background Removal with Edge Attention, CLAHE, and Canny edge detection, followed by ROI extraction with YOLOv7. The classification model was built using RegNetY002 architecture and evaluated via 10-fold stratified cross-validation. The system achieved 98% accuracy and specificity for Type 2 vs. Type 1/A-ROP discrimination, 90% accuracy for Stage 2 vs. Stage 3, and 91% accuracy for A-ROP vs. Type 1 ROP, with

AUC values of 0.98, 0.85, and 0.91, respectively. The study demonstrates that deep learning can effectively distinguish ROP subtypes and reduce ophthalmologists' workload in screening and management. However, the authors acknowledge limitations, including the small dataset, retrospective single-center design, lack of stages 4–5 data, and potential overfitting due to limited training samples. They recommend expanding datasets, improving image quality, and performing multicenter validation before clinical deployment.

The paper titled "Early Detection of Retinopathy of Prematurity Stage Using Deep Learning Approach" by Anand Vinekar, Keerthi Ram et al. [24] presents a Mask R-CNN–based framework for detecting the ridge or demarcation line—a key indicator of Stage 1 and Stage 2 ROP—in neonatal retinal fundus images. They study used 220 retinal images from 45 infants obtained from the KIDROP project, with manually annotated ridge regions for supervised training. To address poor image quality typical of neonatal fundus images, the authors applied modified histogram equalization (MHE) and adaptive CLAHE on the lightness component in YIQ color space, followed by Wiener filtering to reduce noise and enhance the demarcation line. The Mask R-CNN model, implemented with a ResNet-101-FPN backbone, localized ridge regions with an accuracy of 0.88, precision of 0.97, and recall of 0.90, outperforming traditional watershed and vessel enhancement methods. The study demonstrated that pre-processing significantly improves ridge localization, enabling more reliable early-stage ROP detection. However, limitations include a small dataset, single imaging source, and difficulty in distinguishing Stage 1 from Stage 2 ROP due to subtle visual differences. The authors propose extending the approach for full ROP staging and segmentation using larger multi-center datasets for clinical translation.

2.3 Gap Analysis

From the comprehensive review of existing literature on ROP detection using AI and DL techniques, several research gaps and limitations have been identified. These gaps highlight the need for an improved and more practical solution.

1. Lack of Robust Stage-wise Classification: Most existing systems focus only on binary classification (ROP vs non-ROP) and fail to provide accurate stage-wise classification of the disease. This limits their usefulness in clinical decision-making and treatment planning.

2. Dependence on High-Cost Imaging Systems: Many models are trained on high-resolution images captured using expensive devices such as RetCam. This restricts their applicability in low-resource healthcare settings where such equipment is not available.
3. Limited Dataset Size and Diversity: A significant number of studies rely on small, single-centre datasets. This leads to overfitting and reduces the generalization capability of models when applied to new populations or different imaging conditions.
4. Poor Handling of Low-Quality Images: Neonatal retinal images often suffer from noise, low illumination, and motion blur. Existing systems do not effectively address these issues, resulting in decreased model performance.
5. Lack of Explainability in AI Models: Many deep learning models act as “black boxes,” providing predictions without clear reasoning. This lack of interpretability reduces trust among clinicians and limits real-world adoption.
6. Absence of Integrated Clinical Data: Most image-based models do not incorporate important clinical parameters such as gestational age, birth weight, and oxygen exposure, which are critical for accurate ROP prediction.
7. Limited Feature Localization and Segmentation: Although some studies use segmentation techniques, many systems fail to accurately highlight key diagnostic features such as demarcation lines, ridges, and vascular abnormalities.
8. Lack of Real-Time and User-Friendly Applications: Many proposed models remain at the research stage and are not integrated into practical systems or user interfaces for real-time clinical usage.
9. Inadequate Cross-Dataset Validation: Existing models are often not tested on external datasets, leading to poor performance when applied in different clinical environments or populations.
10. High Computational Requirements: Advanced models such as Vision Transformers and hybrid architectures require high computational resources, making them unsuitable for deployment in real-time clinical settings.

CHAPTER 3

SYSTEM ANALYSIS & DESIGN

3.1 Architecture Diagram

3.1.1 Architecture Diagram

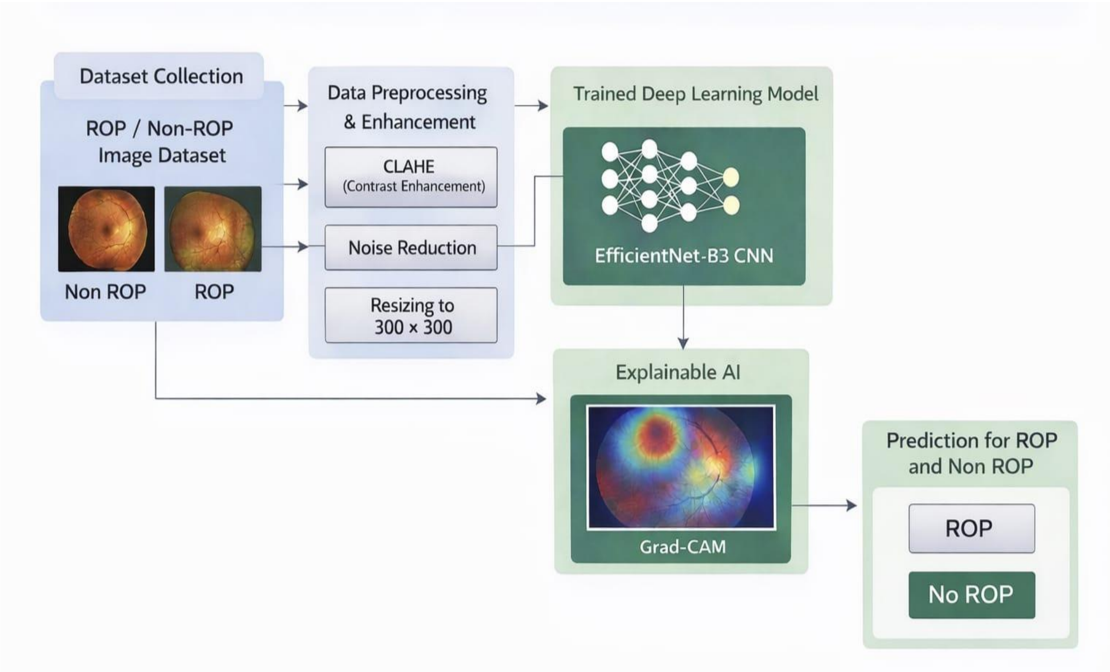

Figure 3.1: System Architecture

The architecture of the proposed system is designed to perform automated detection of ROP using DL techniques. It consists of multiple interconnected modules that process retinal images from input to final prediction and visualization.

The system begins with dataset collection, which includes retinal fundus images categorized into ROP and non-ROP classes. These images are then passed through a data preprocessing and enhancement module, where techniques such as CLAHE, noise reduction, and resizing to a fixed dimension (300 × 300) are applied to improve image quality and feature visibility.

The preprocessed images are fed into a trained DL model based on EfficientNet architecture, which extracts important features and performs classification. The model predicts whether the input image belongs to the ROP or non-ROP category.

To improve interpretability, an XAI module using Grad-CAM is integrated into the system. This module generates a heatmap that highlights the regions of the retinal image that contributed most to the model's prediction.

Finally, the system outputs the prediction result (ROP / non-ROP) along with visual explanations, enabling better understanding and clinical decision-making.

3.1.2 Workflow Description

The workflow of the proposed system follows a sequential pipeline:

1. Image Input:
   Retinal fundus images are collected and provided as input to the system.

2. Preprocessing and Enhancement:
   The input images undergo preprocessing steps including:

   - CLAHE for contrast enhancement
   - Noise reduction using Gaussian blur
   - Resizing to 300 × 300 pixels

3. Model Processing:
   The enhanced image is passed to the trained EfficientNet-based CNN model, which extracts features and performs classification.

4. Prediction Generation:
   The model produces a probability score, which is used to classify the image as:

   - ROP
   - Non-ROP

5. Explainability (Grad-CAM):
   Grad-CAM generates a heatmap highlighting important regions influencing the prediction.

6. Output Display:
   The system displays:

   - Predicted class

- Confidence score
- Heatmap visualization

7. Report Generation
   A PDF report is generated containing prediction results and visual explanations.

3.1.3 Data Flow

The data flow in the system can be described as follows:

Input Layer

- Retinal fundus image (ROP / non-ROP)

Processing Layer

- Image validation
- Image enhancement (CLAHE, noise reduction)
- Image resizing and normalization
- Feature extraction using CNN

Model Layer

- EfficientNet deep learning model
- Probability-based classification

Explainability Layer

- Grad-CAM heatmap generation

Output Layer

- Prediction result (ROP / non-ROP)
- Confidence score
- Visual heatmap
- PDF report generation

3.2 System Design

3.2.1 Design Approach

The proposed system follows a modular and layered design approach, ensuring scalability, maintainability, and ease of integration. The architecture is divided into distinct functional modules, each responsible for a specific task in the ROP detection pipeline.

The system adopts a pipeline-based design, where data flows sequentially from input to output through multiple processing stages. This includes image validation, preprocessing, feature extraction, classification, explainability, and report generation.

The design is centred around the following principles:

- Modularity: Each component such as preprocessing, prediction, and visualization operates independently, allowing easy updates and maintenance.
- Scalability: The system can be extended to include multi-class classification (stage-wise detection) or additional datasets.
- User-Centric Design: A web-based interface ensures ease of use for clinicians and healthcare professionals.
- Explainability: Integration of Grad-CAM ensures transparency in predictions, improving trust in the system.
- Efficiency: Lightweight processing and optimized model inference enable near real-time predictions.

3.2.2 UML Diagrams (Use Case, Class)

3.2.1 Use Case Diagram

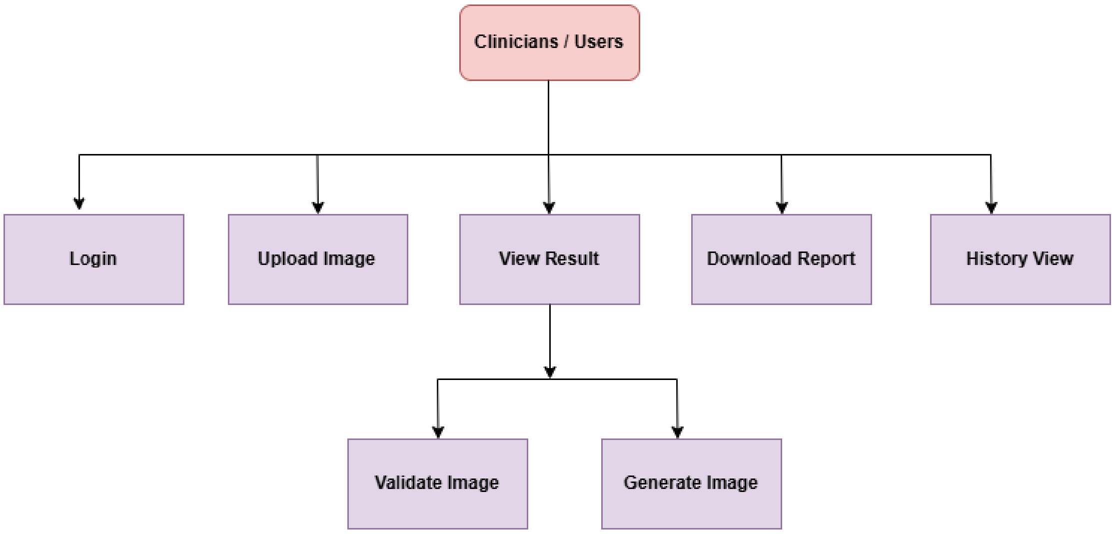

Figure 3.2: Use Case diagram

Actor:

- Clinician / User

Use Cases:

- Login to system
- Upload retinal image
- Validate image

View prediction result

- View Grad-CAM visualization
- Download diagnostic report
- View history records

This diagram illustrates how the user interacts with the system to perform ROP detection.

3.2.1 Class Diagram

Main Classes in Your System: Flask App, ImageProcessor, ModelHandler, GradCAM, ReportGenerator, HistoryManager.

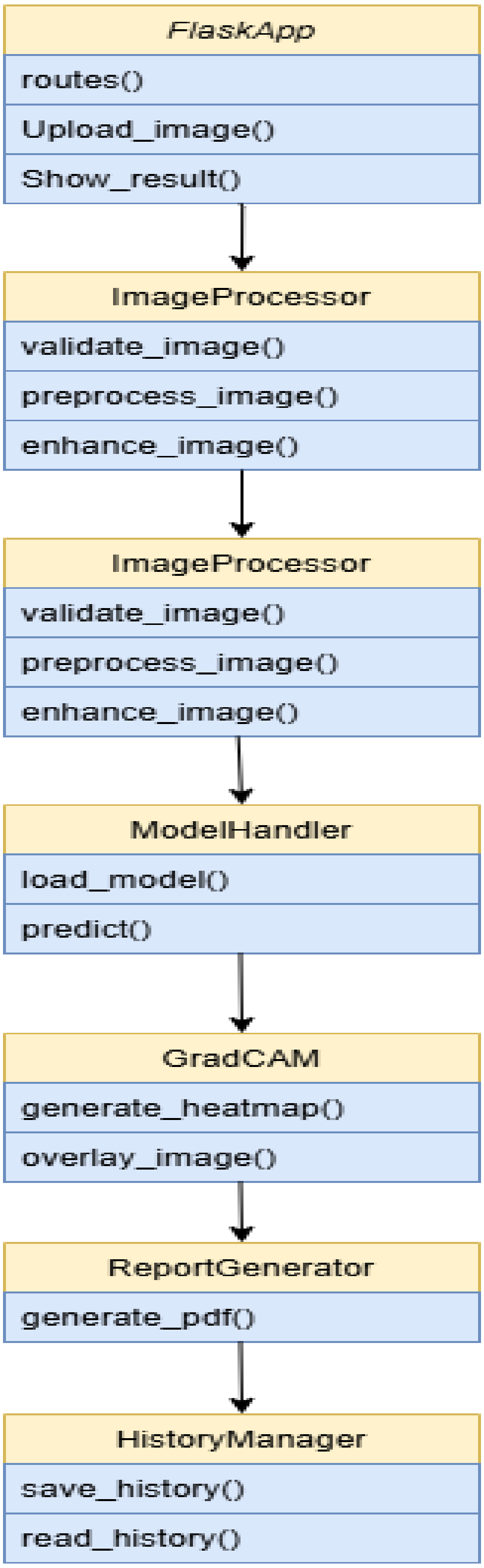

Figure 3.3: Class Diagram

3.3 Module Description

The proposed system is divided into several functional modules, each responsible for a specific task in the ROP detection pipeline. These modules work together to ensure accurate prediction, explainability, and user interaction.

1. Image Upload and Validation Module

This module allows the user (clinician) to upload retinal fundus images through the web interface. Once the image is uploaded, the system verifies whether it is a valid retinal image.

Validation is performed using grayscale analysis, where the number of dark pixels (representing retinal borders) is calculated. If the image does not meet the threshold criteria, it is classified as an invalid image, and an error message is displayed to the user.

Functions involved:

- Image upload via Flask
- is_retinal_image() function

2. Image Preprocessing Module

The preprocessing module enhances the quality of the input retinal image to improve model performance. Since neonatal retinal images often suffer from poor illumination and noise, multiple enhancement techniques are applied.

Key preprocessing steps include:

- Gaussian Blur: Reduces noise and smoothens the image
- CLAHE: Improves contrast and highlights retinal structures
- Color Space Conversion (RGB → LAB): Enhances luminance processing
- Resizing: Standardizes image size to 300 × 300 pixels
- Normalization: Uses EfficientNet preprocessing

This module ensures that the input image is optimized for feature extraction.

Functions involved:

- enhance_fundus_image()
- preprocess_image()

3. Feature Extraction and Prediction Module

This module is the core component of the system, responsible for analyzing the retinal image and predicting the presence of ROP.

A pre-trained DL model (rop_detection_model.h5) based on EfficientNet architecture is used. The model extracts relevant features such as blood vessel patterns and retinal abnormalities and outputs a probability score.

Based on the predicted probability:

- If probability > 0.7 → Classified as ROP
- Else → Classified as NON-ROP

The confidence score is also calculated and displayed.

Functions involved:

- Model loading (load_model)
- model.predict()

4. Explainability Module (Grad-CAM)

To improve transparency and trust, the system integrates Grad-CAM.

This module generates a heatmap highlighting the regions of the retinal image that influenced the model's prediction. It uses the last convolutional layer of the model to compute gradients and visualize important features.

The heatmap is overlaid on the original image for better interpretation.

Functions involved:

- make_gradcam_heatmap()
- save_gradcam()

5. Report Generation Module

This module generates a detailed diagnostic report in PDF format using the ReportLab library.

The report includes:

- Prediction result (ROP / non-ROP)
- Confidence score
- Uploaded retinal image
- Grad-CAM heatmap
- Explanation of results

The report is automatically saved and made available for download.

Functions involved:

- generate_report()

6. Web Interface Module

The system includes a user-friendly web interface developed using Flask and HTML templates.

Features:

- Image upload page
- Result display page
- Grad-CAM visualization
- Report download option
- History viewing

This module ensures ease of use for clinicians and enables real-time interaction.

Files involved:

- index.html
- result.html

- history.html

7. History Management Module

This module stores the prediction results for future reference and analysis.

Each prediction is saved in a CSV file along with:

- Image name
- Prediction result
- Confidence score
- Timestamp

Users can view past predictions through the history page.

Functions involved: save_history(), read_history()

CHAPTER – 4

IMPLEMENTATION

4.1 Technologies Used

The proposed system is developed using a combination of machine learning, image processing, and web development technologies. These technologies enable efficient data processing, model prediction, and user interaction through a web-based interface.

4.1.1 Software Tools

The following software tools were used for the development and implementation of the system:

- Python: The primary programming language used for implementing the machine learning model, image processing, and backend logic.
- Jupyter Notebook: Used for model development, experimentation, training, and testing of the deep learning model.
- Visual Studio Code (VS Code): Used as the development environment for building the Flask web application and integrating different modules.
- Flask: A lightweight web framework used to develop the backend of the web application and handle user requests.
- GitHub: Used for version control and project management.

4.1.2 Hardware Requirements

The system requires the following hardware configuration for efficient execution:

Minimum Requirements:

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB
- Storage: 256 GB

Recommended Requirements:

- Processor: Intel Core i7 or higher

- RAM: 16 GB
- GPU: NVIDIA GPU (for faster model training and inference)
- Storage: 512 GB or higher

The system can run on standard computing devices, but GPU support significantly improves performance during model training.

4.1.3 Frameworks / Libraries

The implementation of the system utilizes several frameworks and libraries for machine learning, image processing, and web development:

- TensorFlow & Keras: Used for building, training, and loading the deep learning model (EfficientNet-based CNN).
- OpenCV (cv2): Used for image processing tasks such as resizing, noise reduction, color space conversion, and CLAHE enhancement.
- NumPy: Used for numerical computations and array manipulations.
- EfficientNet (Keras Applications): Used as the base architecture for feature extraction and classification.
- Flask: Handles routing, request processing, and web application functionality.
- ReportLab: Used for generating PDF reports containing prediction results and visualizations.
- CSV Module: Used for storing and managing prediction history.

4.2 Algorithms / Models:

4.2.1 Algorithm Description

The proposed system implements a deep learning-based pipeline for automated detection of ROP from retinal fundus images. The algorithm integrates image preprocessing, feature extraction using an EfficientNet-based CNN, classification, explainability using Grad-CAM, and report generation.

Step-by-Step Algorithm

Step 1: Input Acquisition

The system begins by accepting a retinal fundus image uploaded by the clinician through a web-based interface.

The uploaded image is stored in the server directory for further processing.

- Input format: RGB image
- Source: Retinal fundus camera images
- Storage path: /static/uploads/

Step 2: Image Validation

To ensure the correctness of input data, the system verifies whether the uploaded image is a valid retinal image.

- The image is resized to 300 × 300 pixels
- Converted to grayscale
- Pixel intensity analysis is performed
- The number of dark pixels (intensity < 30) is calculated

Decision Rule:

- If dark pixel count > threshold → Valid retinal image
- Else → Invalid image

This step prevents non-retinal or corrupted images from being processed further.

Step 3: Image Preprocessing and Enhancement

Preprocessing is crucial to improve image quality and enhance important retinal features.

3.1 Noise Reduction

- Gaussian Blur is applied using a 5×5 kernel
- Reduces high-frequency noise and smoothens the image

3.2 Color Space Transformation

- Image is converted from RGB to LAB color space
- The L-channel (luminance) is separated for enhancement

3.3 Contrast Enhancement (CLAHE)

- CLAHE is applied to the L-channel
- Improves local contrast and enhances blood vessel visibility
- Prevents over-amplification of noise

3.4 Image Reconstruction

- Enhanced L-channel is merged with A and B channels
- Converted back to RGB format

3.5 Resizing

- Image is resized to 300 × 300 pixels
- Ensures compatibility with model input dimensions

3.6 Normalization

- EfficientNet preprocessing is applied
- Pixel values are scaled appropriately

Step 4: Feature Extraction using CNN

The preprocessed image is passed into a trained EfficientNet-based Convolutional Neural Network.

- EfficientNet uses compound scaling (depth, width, resolution)
- Extracts hierarchical features:
    - Low-level: edges, textures
    - Mid-level: vessel structures
    - High-level: pathological patterns

The output of this stage is a feature representation of the retinal image.

Step 5: Prediction

The model processes the extracted features and outputs a probability score:

- $p \in [0,1]$
- Represents likelihood of ROP

Example:

- $p = 0.85$→ High probability of ROP
- $p = 0.30$→ Low probability of ROP

Step 6: Classification Rule

A threshold-based decision mechanism is used:

- If $p > 0.7$→ ROP
- Else → NON-ROP

This threshold ensures higher confidence in positive predictions and reduces false positives.

Step 7: Confidence Calculation

Confidence indicates the certainty of prediction:

$$\text{Confidence} = \max(p, 1 - p)$$

- If prediction = ROP → confidence = p
- If prediction = Non-ROP → confidence = $1 - p$

The value is converted into percentage form for display.

Step 8: Explainability using Grad-CAM

To enhance interpretability, Grad-CAM is applied:

- Uses gradients of the final convolutional layer
- Identifies important regions influencing prediction
- Generates a heatmap

Process:

1. Compute gradients of output with respect to feature maps
2. Perform global average pooling
3. Multiply weights with feature maps
4. Apply ReLU activation
5. Normalize heatmap

The heatmap is superimposed on the original image for visualization.

Step 9: Output Generation

The system generates and displays the following outputs:

- Predicted class (ROP / NON-ROP)
- Confidence score (%)
- Grad-CAM heatmap visualization

Additionally, a PDF report is generated containing:

- Prediction result
- Confidence level
- Uploaded retinal image
- Grad-CAM visualization
- Explanation of results

Step 10: Data Storage

The prediction details are stored in a CSV file:

- Image name
- Prediction
- Confidence
- Timestamp

This allows tracking and reviewing past results.

4.2.2 Flowchart

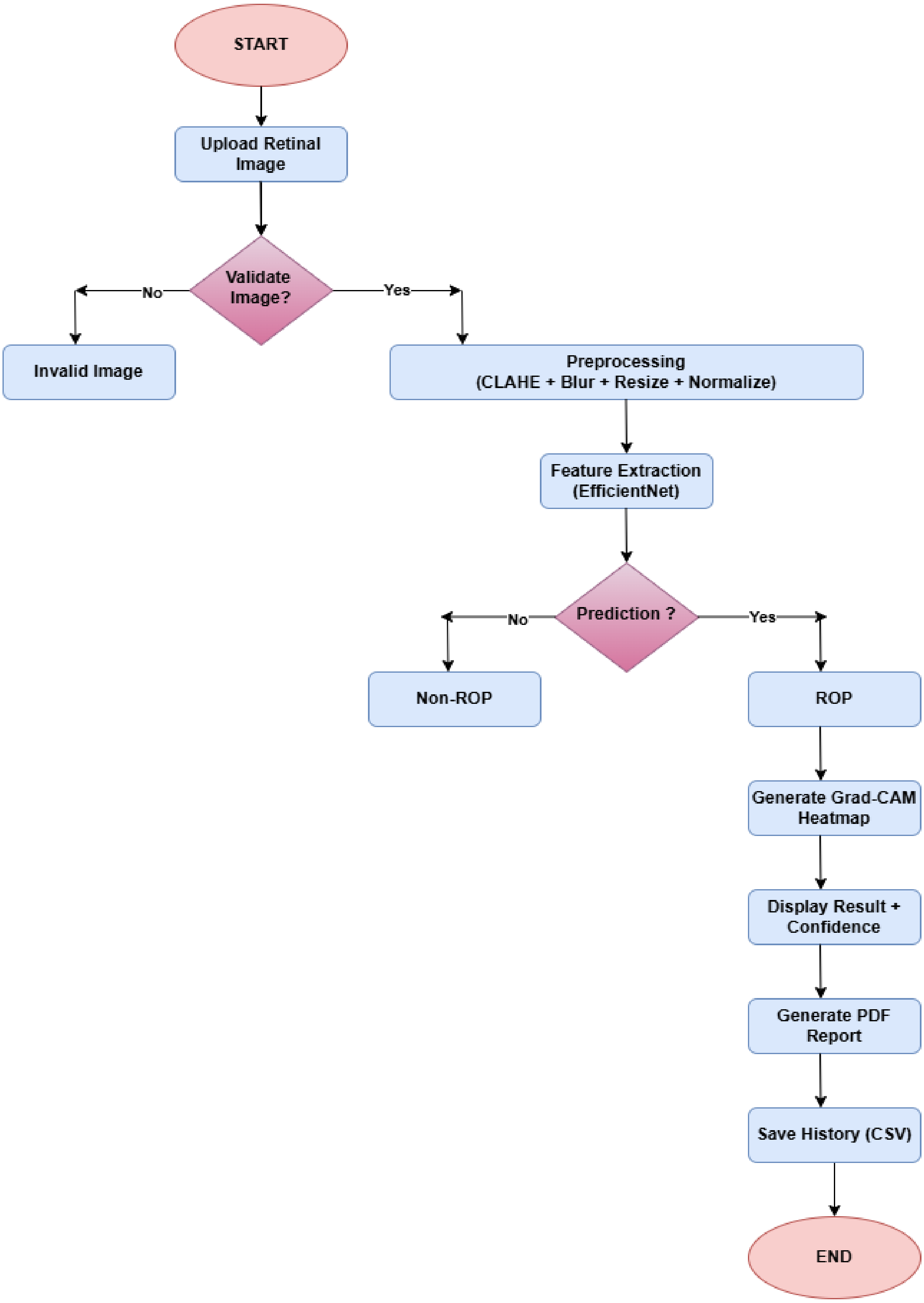

Figure 4.1: Flow of Implementation

Flow Description

The flowchart represents the sequence of operations in the system:

- Start → Upload retinal image
- Check if image is valid
    - If invalid → display error
    - If valid → preprocess image
- Extract features using EfficientNet
- Generate prediction probability
- Apply threshold for classification
- Generate Grad-CAM heatmap
- Display result and generate report
- Save history → End

4.2.3 Mathematical Model

The mathematical formulation of the system can be described as follows:

Input Representation

Let:

- Xbe the input retinal image
- $P(X)$be the preprocessing function

Feature Extraction

The CNN model extracts features:

$$F = f(P(X))$$

where:

- f represents the EfficientNet model
- F represents extracted feature maps

Prediction Function

The model outputs a probability:

$$p = \sigma(F)$$

where:

- $p \in [0,1]$
- $\sigma$is the activation function (sigmoid for binary classification)

Classification Rule

$$\text{Output} = \begin{cases} \text{ROP}, & \text{if } p > 0.7 \\ \text{NON-ROP}, & \text{otherwise} \end{cases}$$

Confidence Score

$$\text{Confidence} = \max\,(p, 1 - p)$$

Grad-CAM Heatmap

Grad-CAM computes importance weights:

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k}$$

Heatmap:

$$L_{Grad-CAM} = \text{ReLU}\left(\sum_k \alpha_k\, A^k\right)$$

where:

- A^k= feature maps
- y= predicted class score

4.3 Working/Screens

4.3.1 Input Processing

The input processing stage involves collecting and preparing the retinal image for analysis.

Steps Involved:

1. **Image Upload**
   - The user uploads a retinal fundus image through the web interface.
   - The image is stored in the server directory (/static/uploads/).
2. **Image Validation**
   - The system checks whether the uploaded image is a valid retinal image.
   - Grayscale conversion is applied.
   - Dark pixel count is calculated to detect retinal boundaries.
   - If invalid, an error message is displayed.
3. **Initial Preparation**
   - The validated image is passed to the preprocessing module for enhancement.

4.3.2 Execution Process

The execution phase is the core processing stage where the image is analyzed using the trained model.

Steps Involved:

1. **Image Preprocessing**
   - Gaussian blur is applied to remove noise
   - CLAHE enhances contrast and highlights retinal structures
   - Image is converted to LAB color space

- Resized to 300 × 300 pixels
- Normalized using EfficientNet preprocessing

2. **Model Inference**
   - The preprocessed image is passed to the trained EfficientNet model
   - Feature extraction is performed automatically by convolutional layers
3. **Prediction Generation**
   - The model outputs a probability score
   - Threshold-based classification determines:
     - ROP
     - NON-ROP
4. **Grad-CAM Visualization**
   - Generates a heatmap highlighting important regions
   - Heatmap is overlaid on the original image

4.3.2 Output Generation

The output stage presents the final results to the user in an understandable and structured format.

Displayed Outputs:

1. **Prediction Result**
   - The system displays whether the image is:
     - ROP
     - NON-ROP
2. **Confidence Score**
   - Displays the probability of prediction as a percentage
3. **Grad-CAM Visualization**
   - Heatmap highlights regions influencing the decision

    - Improves interpretability and trust

4. **Explanation**
    - Provides a brief textual explanation of the prediction

5. **PDF Report Generation**
    - A downloadable report is generated containing:
        - Prediction
        - Confidence
        - Uploaded image
        - Heatmap visualization

6. **History Storage**
    - Prediction details are saved in a CSV file
    - Can be viewed later through the history page

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Outputs

This chapter presents the results obtained from the proposed ROP detection system and provides a detailed discussion of its performance. The system was evaluated based on its ability to accurately classify retinal fundus images as ROP or non-ROP, along with providing explainable visual outputs using Grad-CAM.

5.1.1 Screenshots

Experimental Setup

The system was implemented using Python with TensorFlow and Keras for deep learning, and Flask for web deployment. The model was trained using retinal fundus images after applying preprocessing techniques such as CLAHE, Gaussian blur, and normalization.

Configuration Details:

- Input Image Size: 300 × 300 pixels
- Model: EfficientNet-based CNN
- Classification Type: Binary (ROP / non-ROP)
- Threshold: 0.7
- Explainability: Grad-CAM

Prediction Output

The system successfully classifies retinal images into:

- ROP (Retinopathy detected)
- NON-ROP (No abnormality detected)

Each prediction is accompanied by a confidence score, which indicates the certainty of the model's decision.

```
test_image_path = "/content/retinopathyofprematurity1.jpg"

predict_single_image(test_image_path, model)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 20s 20s/step
Prediction: ROP
Confidence: 92.53 %
('ROP', 0.9252572655677795)
```

```
test_image_path = "/content/images.jpg"

predict_single_image(test_image_path, model)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 57ms/step
Prediction: No ROP
Confidence: 94.8 %
('No ROP', 0.9479874558746815)
```

Figure 5.1: Model prediction results

Visualization Results (Grad-CAM)

The Grad-CAM module generates heatmaps that highlight important regions in the retinal image influencing the prediction.

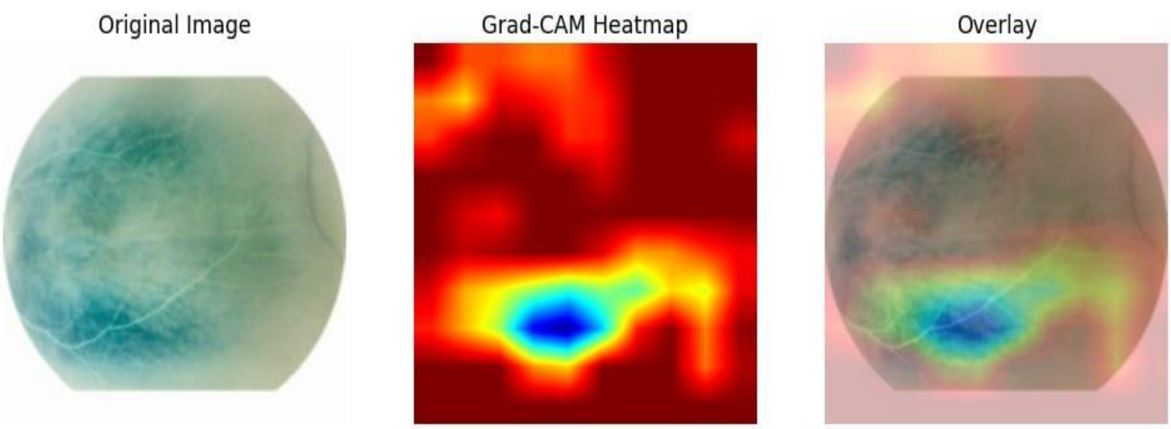

Figure 5.2: Grad-CAM Visualization of ROP Detection

- Red/Yellow regions → High importance

- Blue regions → Low importance

These visualizations help in understanding whether the model is focusing on relevant retinal structures such as blood vessels and abnormal regions.

System Output Interface

The web application displays:

- Uploaded retinal image
- Prediction result
- Confidence score
- Grad-CAM heatmap
- Downloadable PDF report

The system provides real-time results, making it suitable for clinical assistance.

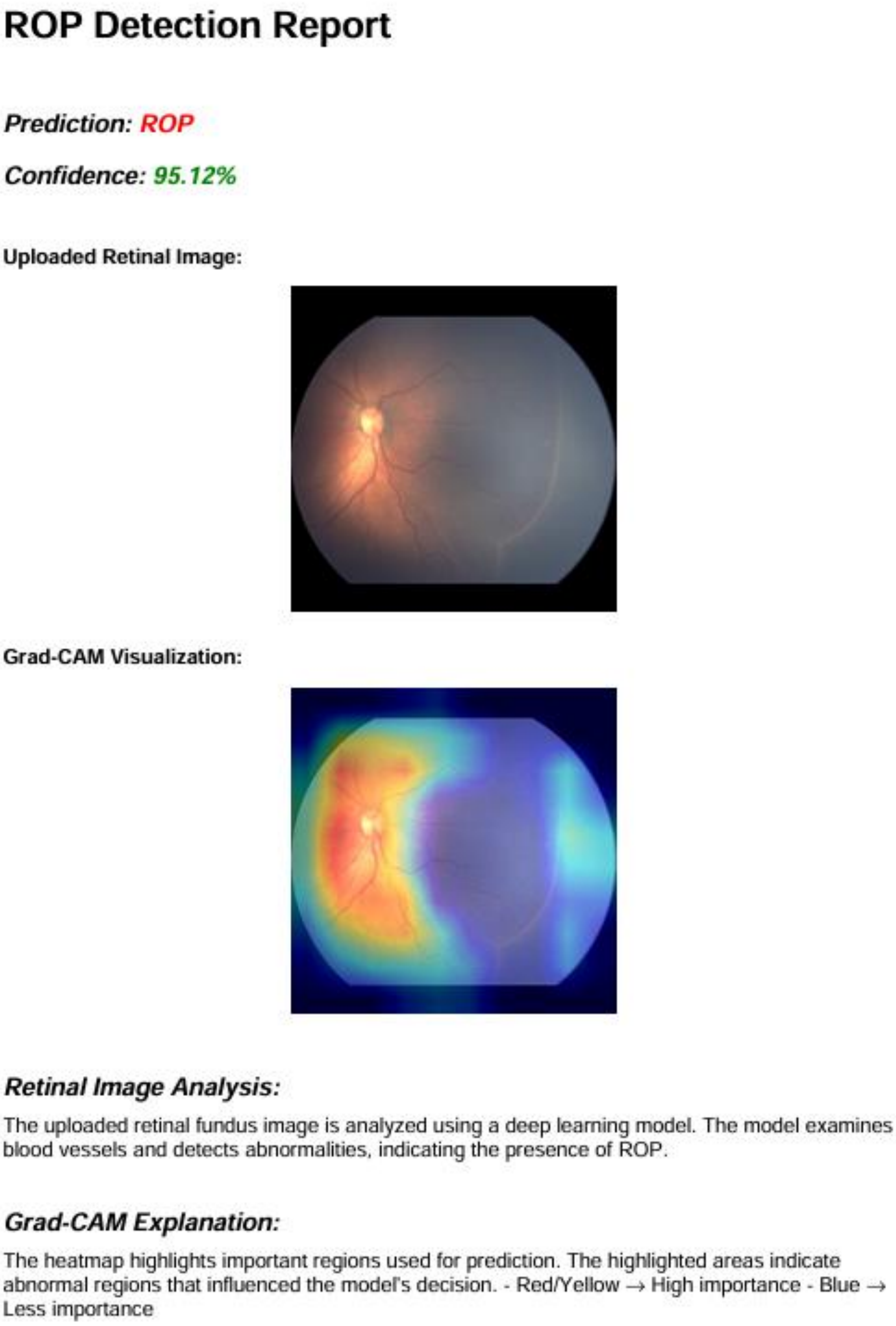

Figure 5.3: Web Interface Output for ROP Detection

5.1.2 Graphical Results

Confusion Matrix Analysis

The performance of the proposed ROP detection model is evaluated using a confusion matrix, which provides a detailed comparison between actual and predicted classifications.

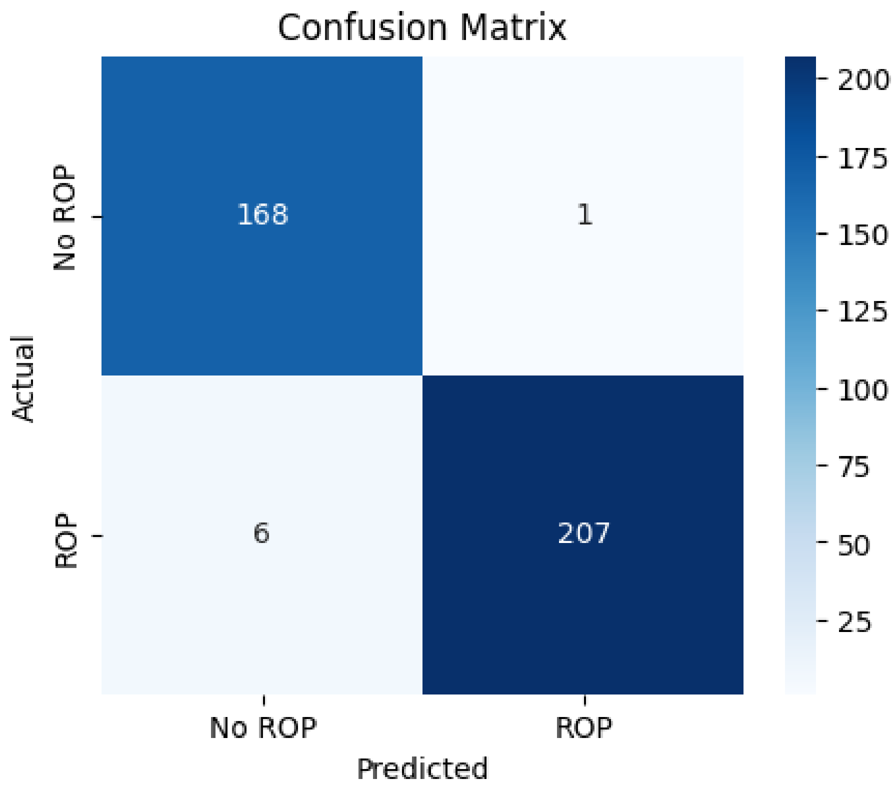

Figure 5.4: Confusion Matrix of ROP Detection Model

	Predicted non-ROP	Predicted ROP
Actual non-ROP	168 (TN)	1 (FP)
Actual ROP	6 (FN)	207 (TP)

5.2 Performance Analysis

5.2.1 Metrics

1. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{207 + 168}{207 + 168 + 1 + 6} = \frac{375}{382} \approx 98.17\%$$

2. Precision (ROP Class)

$$Precision = \frac{TP}{TP + FP}$$

$$= \frac{207}{207 + 1} \approx 99.52\%$$

3. Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

$$= \frac{207}{207 + 6} \approx 97.18\%$$

4. Specificity

$$Specificity = \frac{TN}{TN + FP}$$

$$= \frac{168}{168 + 1} \approx 99.41\%$$

5. F1-Score

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

$$\approx 98.34\%$$

Interpretation of Results

- The model achieves high accuracy (≈98.17%), indicating strong overall performance.
- Very high precision (≈99.52%) shows that the model rarely misclassifies non-ROP as ROP (low false positives).
- High recall (≈97.18%) indicates that most ROP cases are correctly detected.
- High specificity (≈99.41%) confirms reliable identification of non-ROP cases.

- The low number of false negatives (6 cases) is important, as missing ROP cases is critical in medical diagnosis.

Observations

- The model performs exceptionally well in distinguishing between ROP and non-ROP cases.
- The confusion matrix shows a balanced performance across both classes.
- Grad-CAM visualization further supports the correctness of predictions by highlighting relevant retinal regions.
- The system demonstrates clinical reliability and robustness.

The confusion matrix clearly demonstrates that the proposed EfficientNet-based model provides high accuracy, strong sensitivity, and excellent specificity, making it suitable for real-world ROP screening applications.

5.2.2 Evaluation Results:

The model achieves an overall accuracy 98%, which indicates high performance. Both classes have high precision and recall values; it shows that the model performs well in detecting both ROP and NON-ROP.

Table 5.1: Performance Metrics of the Proposed Model

Class	Precision	Recall	F1-Score	Support
No ROP	0.98	0.98	0.98	169
ROP	0.98	0.99	0.98	213
Accuracy			0.98	382

Macro Avg	0.98	0.98	0.98	382

5.3 Comparison

5.3.1 Comparison with Existing Systems

The proposed system is compared with existing ROP detection systems based on key parameters such as accuracy, usability, explainability, and deployment feasibility.

Table 5.2: Comparison of Existing Systems and Proposed System

Feature	Existing Systems	Proposed System
Classification Type	Mostly binary (ROP / Non-ROP)	Binary (ROP / Non-ROP)
Stage-wise Detection	Limited or absent	Not implemented (future scope)
Image Quality Handling	Limited preprocessing	CLAHE + noise reduction
Explainability	Mostly black-box models	Grad-CAM visualization
User Interface	Mostly research-based (no UI)	Web-based interface (Flask)
Real-time Prediction	Limited	Supported
Report Generation	Not available	PDF report generation
Data Storage	Rarely included	History tracking (CSV)
Cost Efficiency	Depends on high-end imaging	Works with 2D images

5.3.2 Advantages of Proposed System

The proposed ROP detection system offers several significant advantages over existing approaches, particularly in terms of accuracy, usability, and practical deployment.

1. Automated and Efficient Diagnosis

The system automates the detection of ROP using deep learning, reducing the dependency on manual examination by ophthalmologists. This helps in minimizing human error and speeds up the screening process.

2. High Prediction Accuracy

The use of an EfficientNet-based CNN enables effective feature extraction and classification, resulting in high accuracy and reliable predictions for ROP detection.

3. Robust Image Preprocessing

The integration of preprocessing techniques such as CLAHE, Gaussian blur, and normalization enhances image quality. This improves visibility of retinal features and increases model performance even with moderately low-quality images.

4. Explainable AI using Grad-CAM

Unlike many traditional deep learning models, the proposed system incorporates Grad-CAM visualization. This highlights important regions in the retinal image that influence the prediction, improving transparency and building trust among clinicians.

5. User-Friendly Web Interface

The system provides a simple and interactive web-based interface developed using Flask. Clinicians can easily upload images, view results, and download reports without requiring technical expertise.

6. Real-Time Prediction Capability

The system processes images quickly and provides predictions in near real-time, making it suitable for clinical environments where timely diagnosis is critical.

7. Automated Report Generation

The system generates structured PDF reports that include prediction results, confidence scores, and Grad-CAM visualizations. This facilitates documentation and easy sharing of diagnostic results.

8. History Management System

All predictions are stored in a CSV file along with timestamps, allowing users to track previous results and maintain records for future reference.

9. Cost-Effective and Accessible Solution

The system works with 2D retinal fundus images, eliminating the need for expensive 3D imaging devices such as RetCam. This makes it suitable for deployment in low-resource healthcare settings.

10. Modular and Scalable Design

The system is designed in a modular manner, allowing future enhancements such as stage-wise classification, integration of clinical data, or deployment on cloud platforms.

5.3 Limitations

Despite the promising performance of the proposed ROP detection system, certain limitations exist that need to be addressed for broader clinical applicability and improved robustness.

Binary Classification Only - The current system performs only binary classification (ROP vs non-ROP). It does not provide detailed stage-wise classification of ROP, which is crucial for determining disease severity and treatment planning.

Dependence on Dataset Quality and Size - The performance of the model is highly dependent on the dataset used for training. Limited dataset size or lack of diversity may lead to reduced generalization when applied to different populations or imaging conditions.

Sensitivity to Image Quality - Although preprocessing techniques are applied, extremely poor-quality images with severe noise, blur, or improper illumination may still affect the accuracy of predictions.

Lack of Clinical Data Integration - The system relies only on image-based inputs and does not incorporate important clinical parameters such as gestational age, birth weight, and oxygen exposure, which are significant factors in ROP diagnosis. The system has been tested primarily in a controlled environment. It has not yet undergone extensive validation across multiple hospitals or real-world clinical settings, which is essential for practical deployment.

Computational Requirements - Although EfficientNet is optimized, the system still requires moderate computational resources for model inference and Grad-CAM generation, which may be a limitation in low-end devices. While Grad-CAM provides visual explanations, it does not fully explain all aspects of the model's decision-making process, and interpretation still requires domain knowledge.

While the proposed system demonstrates strong performance and practical usability, addressing these limitations through improvements such as stage-wise classification, multimodal data integration, larger datasets, and real-world validation will enhance its effectiveness and clinical reliability.

CHAPTER – 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

6.1.1 Summary of Work

The work carried out in this project includes:

- Collection and preprocessing of retinal fundus images
- Implementation of image enhancement techniques such as CLAHE and Gaussian blur
- Development of a deep learning model using EfficientNet architecture
- Integration of Grad-CAM for explainable predictions
- Design and development of a Flask-based web application
- Implementation of real-time prediction and visualization
- Generation of automated PDF reports
- Storage and management of prediction history

The system successfully performs automated ROP detection with high accuracy and provides meaningful visual insights.

6.1.2 Advantages

The key advantages of the proposed system include:

- Automated and efficient detection of ROP
- High accuracy and reliable predictions
- Explainable AI through Grad-CAM visualization
- User-friendly web interface
- Real-time prediction capability
- Automated report generation

- Cost-effective solution using 2D images
- Easy deployment and scalability

6.1.3 Applications

The proposed system can be applied in various real-world scenarios:

- Neonatal Intensive Care Units (NICUs): For early screening of premature infants
- Hospitals and Eye Clinics: Assisting ophthalmologists in diagnosis
- Telemedicine Platforms: Remote diagnosis in rural or underserved areas
- Medical Training: Educational tool for students and trainees
- Research Applications: Supporting further studies in ophthalmology and AI

6.1.4 Limitations

Despite its effectiveness, the system has certain limitations:

- Performs only binary classification (ROP / non-ROP)
- Depends on image quality for accurate predictions
- Limited dataset may affect generalization
- Does not incorporate clinical parameters
- Requires further validation in real-world clinical settings

6.2 Future Scope

The proposed system can be further enhanced and extended in several ways:

1. Stage-wise Classification of ROP: Extend the model to classify different stages of ROP for better clinical decision-making.
2. Integration of Clinical Data: Incorporate patient-specific parameters such as birth weight, gestational age, and oxygen exposure to improve prediction accuracy.
3. Larger and Diverse Dataset: Train the model on multi-center datasets to improve generalization and robustness.

4. Mobile and Cloud Deployment: Develop mobile applications or cloud-based platforms for wider accessibility and real-time remote diagnosis.
5. Advanced Explainable AI Techniques: Integrate additional XAI methods to provide deeper insights into model decisions.
6. Real-Time Screening System: Optimize the model for faster inference to enable real-time screening in clinical environments.
7. Integration with Hospital Systems: Connect the system with electronic health records (EHR) for seamless data management.

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APPENDIX A

Front Page of the Conference Paper

Explainable Deep Learning Framework for Early Risk Prediction of Retinopathy of Prematurity Using EfficientNet-B3 and Grad-CAM

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Abstract—**Retinopathy of Prematurity (ROP) is an extreme retinal disease that impacts premature infants and might result in permanent vision loss if not detected early. Timely screening and analysis are crucial for stopping extreme complications. However, manual screening requires skilled ophthalmologists and can be time consuming in large-scale neonatal screening programs.**

This paper proposes a deep learning-based predictive framework for computerized detection of ROP using retinal fundus images. The proposed system makes use of the EfficientNet-B3 Convolutional Neural Network (CNN) structure to extract discriminative capabilities from retinal images and classify them into ROP and Non-ROP categories. The dataset used in this study includes 2542 retinal fundus images, which include 1417 ROP images and 1125 non-ROP images, with all images resized to a resolution of 300×300 pixels. The dataset is split into training, validation, and testing units using a stratified sampling method.

To enhance version interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is included to focus on the retinal areas that affect the version's predictions. Experimental outcomes exhibit that the proposed version achieves a standard accuracy of 98%, with excessive precision and consider values for each class. The low range of fake negatives shows the reliability of the version for clinical screening applications.

The proposed method can help ophthalmologists in early detection of ROP and might serve as a supportive system in neonatal screening programs, supporting lessen diagnostic workload and enhance medical decision-making.

Keywords—Retinopathy of Prematurity, Deep Learning, EfficientNet-B3, Medical Image Analysis, Explainable AI, Grad-CAM

I. Introduction

ROP is a vasoprolliferative retinopathy that happens to premature infants because of the improper development of vascularity. Unless early identified, it may cause serious visual problems, such as retinal detachment. Global research states that ROP is still a prevalent source of morbidity in premature babies with a combined prevalence of 31.9 and more often seen in low and middle-income nations, thus the significance of early identification and screening interventions [2]. An example retinal fundus image is shown in Fig. 1.

The disease has five stages starting with mild abnormalities up to full retinal detachment and thus it is critical to detect the disease at an earlier stage. Screening and following the progression of the disease is important to fundus imaging. Even though Artificial Intelligence (AI) has demonstrated potential in automated ROP diagnosing, it has not been fully used in clinical practice because of the absence of a variety of retinal data of infants in large quantities. Thus, it is essential to create effective AI models based on well-annotated fundus images to enhance the quality of reliable and accessible ROP diagnosis [3]

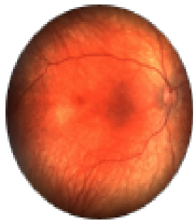

Fig. 1. Sample retinal fundus image from the publicly available Kaggle dataset used in this study.

The low contrast, noise, and uneven illumination of neonatal fundus images commonly affect (ROP) diagnosis. To solve this problem, enhancing methods of images, including Contrast Limited Adaptive Histogram Equalization (CLAHE) and Automated Multi-Scale Retinex with Color Restoration (MSRCR) have been researched as a way of enhancing image clarity. Such techniques make the retinal structures much more visible, particularly when dealing with low-quality images, which helps in making more accurate clinical assessment. The research also notes the significance of making such improvement methods a part of clinical processes by using user-friendly interfaces, which facilitates a more effective and credible ROP diagnosis [5].

APPENDIX B

Conference Submission Letter

Submissions Contact Chairs Help Center Select Your Role : **Author** ICICACS2026 Pravallika Chenna

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Paper ID	1506
Paper Title	Explainable Deep Learning Framework for Early Risk Prediction of Retinopathy of Prematurity Using EfficientNet-B3 and Grad-CAM

Reviewer #3

Questions

1. Review Comments

Topic: Very useful

Abstract: Acceptable

Literature Review: There are 27 papers listed in the reference list at the end. But the authors have not referred/cited the papers: 4, 6, 8, 12, 16, 17, 18, 19, 20, 22, 23, 25, 26, 27. This needs some correction in the review section.

Methodology: Overall approach is good.

Experimental Results: Since only one set of data (samples 2000+) is used in this work, the reliability of the performance metrics cannot be accepted as it is.

Presentation: Overall is Above Average.

NOTE: The figures (1, 2 and 3) are used from some other sources. This can be done only with prior permission from the original sources/authors of the document/paper/book.

This correction is very very essential, otherwise this figures may be considered as plagiarism.

APPENDIX C

Source Code

```
import csv
from datetime import datetime
from flask import Flask, render_template, request, send_file
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import load_model
from tensorflow.keras.applications.efficientnet import preprocess_input
import cv2
import os

from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Image
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib.pagesizes import A4
from reportlab.lib.units import inch

app = Flask(__name__)

# Load Model
model = load_model("rop_detection_model.h5")

UPLOAD_FOLDER = "static/uploads"
GRADCAM_FOLDER = "static/gradcam"

app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER
app.config["GRADCAM_FOLDER"] = GRADCAM_FOLDER
```

```
# ========================================
# Image Enhancement (From Notebook)
# ========================================
def enhance_fundus_image(image_path, img_size=300):
    img = cv2.imread(image_path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    img = cv2.resize(img, (img_size, img_size))

    img_blur = cv2.GaussianBlur(img, (5, 5), 0)

    lab = cv2.cvtColor(img_blur, cv2.COLOR_RGB2LAB)
    l, a, b = cv2.split(lab)

    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
    l_clahe = clahe.apply(l)

    lab_clahe = cv2.merge((l_clahe, a, b))
    enhanced_img = cv2.cvtColor(lab_clahe, cv2.COLOR_LAB2RGB)

    enhanced_img = enhanced_img.astype("float32")

    return enhanced_img
```

```python
# ======================================
# Preprocess Image
# ======================================
def is_retinal_image(image_path):
    img = cv2.imread(image_path)
    img = cv2.resize(img, (300, 300))

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Count dark pixels (retina images have black borders)
    dark_pixels = np.sum(gray < 30)

    if dark_pixels > 5000:
        return True
    else:
        return False

def preprocess_image(path):
    img = enhance_fundus_image(path)

    img = preprocess_input(img)

    img = np.expand_dims(img, axis=0)

    return img
```

```python
# =================================
# Grad-CAM
# =================================
def make_gradcam_heatmap(img_array, model, last_conv_layer_name="top_conv"):
    grad_model = tf.keras.models.Model(
        [model.inputs],
        [model.get_layer(last_conv_layer_name).output, model.output]
    )

    with tf.GradientTape() as tape:
        conv_outputs, predictions = grad_model(img_array)
        class_channel = predictions[0]

    grads = tape.gradient(class_channel, conv_outputs)

    pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))

    conv_outputs = conv_outputs[0]

    heatmap = tf.matmul(conv_outputs, pooled_grads[..., tf.newaxis])
    heatmap = tf.squeeze(heatmap)

    heatmap = tf.maximum(heatmap, 0)
    heatmap /= tf.reduce_max(heatmap) + 1e-8

    return heatmap.numpy()
```

```python
# ================================
# Prediction
# ================================
@app.route("/predict", methods=["POST"])
def predict():

    file = request.files["image"]

    filepath = os.path.join(app.config["UPLOAD_FOLDER"], file.filename)
    file.save(filepath)

    # ✅ Step 1: Check if retinal image
    if not is_retinal_image(filepath):
        return render_template("result.html",
                               prediction="Invalid Image",
                               confidence=0,
                               image=filepath,
                               gradcam=None,    # ✅ IMPORTANT
                               explanation="The uploaded image is not a retinal fundus image. \
                                Please upload a valid retinal image.")

    # ✅ Step 2: Preprocess + Predict
    img = preprocess_image(filepath)
    prediction = model.predict(img)


def predict():
    # ✅ Step 3: DEFINE prob (THIS WAS MISSING)
    prob = float(prediction[0][0])

    # ✅ Step 4: Classification
    if prob > 0.7:
        result = "ROP"
    else:
        result = "NON-ROP"

    confidence = prob if prob >= 0.5 else 1 - prob
    confidence = round(confidence * 100, 2)

    # GradCAM
    heatmap = make_gradcam_heatmap(img, model)
    gradcam = save_gradcam(filepath, heatmap)

    # Generate report
    generate_report(result, confidence, filepath, gradcam)

    save_history(file.filename, result, confidence)
```

```python
# Explanation
if result == "ROP":
    explanation = (
        "The retinal image shows abnormal blood vessel patterns, indicating possible "
        "Retinopathy of Prematurity (ROP). The highlighted regions in Grad-CAM "
        "represent areas of concern."
    )
else:
    explanation = (
        "The retinal image appears normal with well-defined blood vessels. "
        "No significant abnormalities are detected, indicating a non-ROP condition."
    )
return render_template("result.html",
                       prediction=result,
                       confidence=confidence,
                       image=file_path,
                       gradcam=gradcam,
                       explanation=explanation)
```

APPENDIX D

Plagiarism Report

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